# Trainings

# Select Service Tools

### Recommended Cummins® Service Tools

- Bore dial gauge, Part Number 3376619

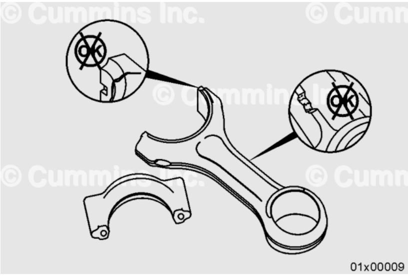
### Additional Service Items

- Vise with brass jaws
- Magnetic particle testing machine
- Ultraviolet light.

# General Information

**CAUTION**

**To prevent damage to the fracture-split connecting rod, do not set the connecting rod or rod cap on the fracture-split connection. Polishing and damage to the mating surface will result.**




The connecting rod uses a fracture-split connection between the connecting rod cap and the connecting rod. Connecting rods with a fracture-split surface **must** be treated with caution. The two pieces of the connecting rod can **not** be rubbed together. The mating surfaces will be damaged. Be careful **not** to drop either piece of the connecting rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification, or completely separated.

# Preparatory Steps

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

**⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

**⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

**⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

**⚠ CAUTION ⚠**

**Rotating the engine with the connecting rod caps loose or removed can cause damage to the piston cooling nozzles. Always remove the piston cooling nozzles prior to removing the connecting rod and piston assembly.**

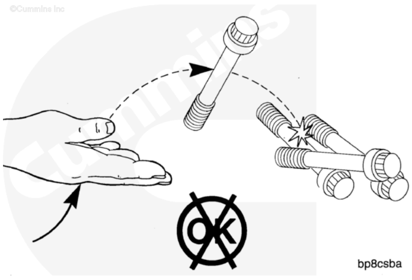
The piston and connecting rod **must** be removed as an assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Remove the piston cooling nozzle. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Remove and disassemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)

## Clean and Inspect for Reuse

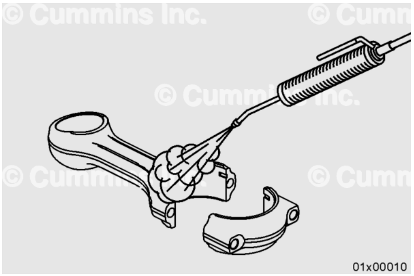
**⚠ CAUTION ⚠**

**Prevent damage to the capscrews. Nicks in the body of the capscrew can cause an area of stress that can malfunction during engine operation. Damage to the threads will cause torque values to be incorrect and will damage the mating parts.**




**⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**




**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

**⚠ WARNING ⚠**

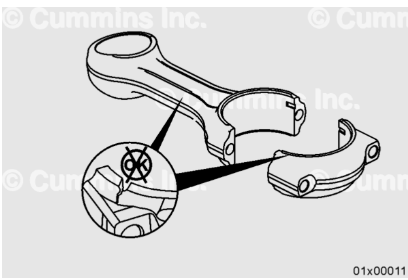
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Use solvent or steam to clean the capscrews, connecting rod, and connecting rod cap.

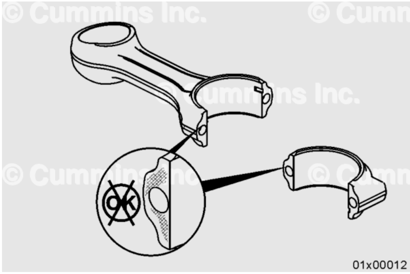
Dry with compressed air.

Inspect the connecting rod and connecting rod cap for damage.

Replace the connecting rod if the I-beam or connecting rod cap is damaged.



Inspect the connecting rod and connecting rod cap for fretting damage on the mating surfaces.



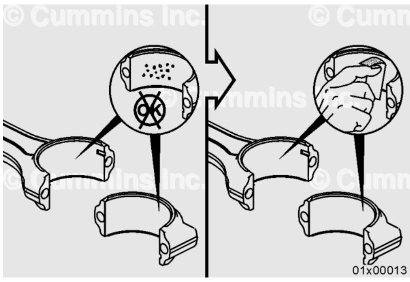

Polished spots on either the connecting rod or connecting rod cap mating surfaces indicate fretting and the connecting rod **must** be replaced.

Dark areas on the face indicate carbonized oil deposits. Connecting rods with dark areas are reusable if there is no fretting present.

The connecting rod and connecting rod cap **must** be replaced as an assembly if fretting damage is visible on either piece.

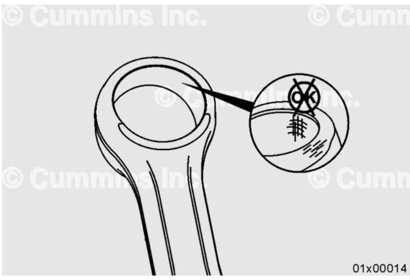
Polishing can be observed on the thrust face of the connecting rod or connecting rod cap, which is normal. Use the following procedure for inspection criteria. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-007.html)

Check the bearing surface for nicks or burrs.




If the nicks or burrs can **not** be removed with a fine emery cloth, the connecting rod **must** be replaced.

Inspect the connecting rod pin bushing for damage.

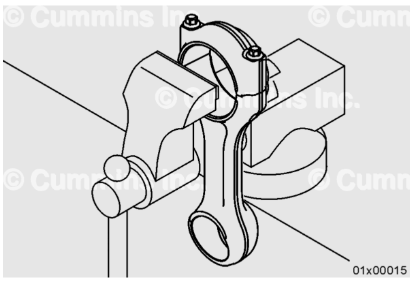



Replace any connecting rod pin bushing with evidence of scoring, galling, or scuffing.

Replace any bushing that has turned in the bore.

Special tools and precision machining are required to replace bushings. If Cummins Inc. approved tools and procedures are **not** available, the connecting rod **must** be replaced.

**⚠ CAUTION ⚠**



**The connecting rod and connecting rod cap have a machined face on one side of the connecting rod. The connecting rod must be installed with the machined surfaces aligned to reduce the possibility of damage to the connecting rods and crankshaft.**

**⚠ CAUTION ⚠**

**Use a vise with brass jaws to hold the connecting rod. Notches, scratches, or dents in the I-beam area can cause engine damage.**

Place the connecting rod into a vise with brass jaws. Support the connecting rod at the large bearing end to avoid I-beam damage.

Install the connecting rod cap onto the connecting rod.

Lubricate the threads of the capscrews. Use clean engine oil.

Thread the capscrews into the connecting rod. Hand tighten.

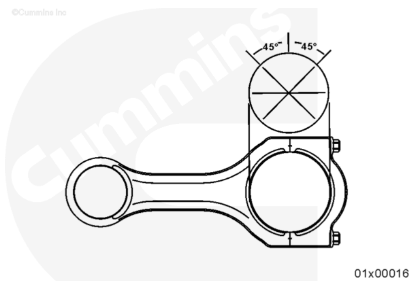
Tighten the capscrews.

**Torque Value:**

1. 91 n•m [ 67 ft-lb ]
2. Rotate 60 degrees.

# Measure

Measure the connecting rod crankshaft bore inside diameter with a bore dial gauge, Part Number 3376619.

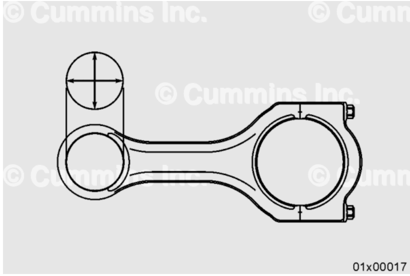



| Connecting Rod Crankshaft Bore Inside Diameter | | |
|---|---|---|
| mm | | in |
| 96.976 | MIN | 3.818 |
| 97.002 | MAX | 3.819 |

If the connecting rod crankshaft bore inside diameter is **not** within specifications, the connecting rod **must** be replaced.

Measure the connecting rod piston pin bushing inside diameter with a bore dial gauge, Part Number 3376619.

| Connecting Rod Piston Pin Bushing Inside Diameter | | |
|---|---|---|
| mm | | in |
| 52.052 | MIN | 2.049 |
| 52.064 | MAX | 2.050 |




If the connecting rod piston pin bushing inside diameter is **not** within specifications, the connecting rod **must** be replaced.

# Magnetic Crack Inspect

This procedure describes the magnetic particle inspection for the connecting rod and connecting rod cap.

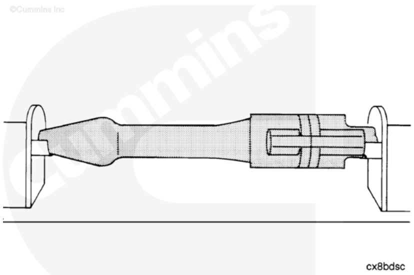
Use a magnetic particle testing machine to check for cracks within the connecting rod.

The connecting rod and connecting rod cap **must** be assembled during this process.

Use the residual method. Apply head shot amperage. Adjust the amperage to 1500 amperes direct current or rectified alternating current.

Check for cracks on the connecting rod.

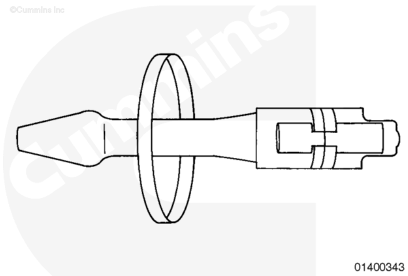
The connecting rod **must** be replaced if any cracks are visible.




Use the residual method. Apply coil shot amperage.

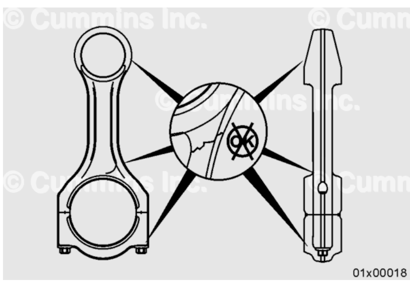
| Amperage (Ampere-Turns) | |
|---|---|
| **Minimum** | **Maximum** |
| 2600 amperes direct current | 2800 amperes direct current |

Ampere-turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.




Check for cracks on the connecting rod.

The connecting rod **must** be replaced if any indications of cracks are visible in the critical (shaded) areas.



**⚠ WARNING ⚠**

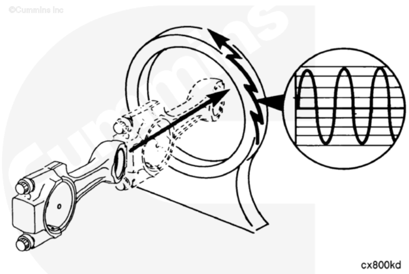
**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**

**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

**⚠ CAUTION ⚠**

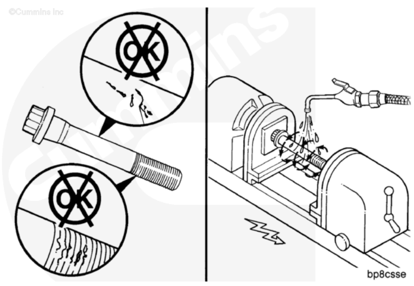
**The connecting rod must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.**




Demagnetize the connecting rod.

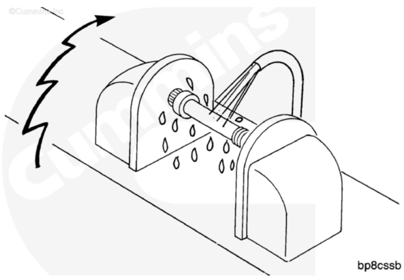
Use solvent or steam to clean the connecting rod.

This procedure describes the magnetic particle inspection for the connecting rod capscrews.




Use a Magnaflux™ or similar magnetic particle testing machine.

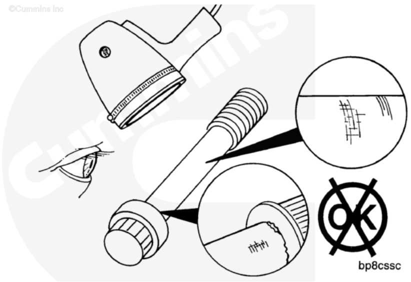
Use the continuous method. Apply a head shot of 300 to 400 amperes direct current or rectified alternating current.




Use an ultraviolet light. Check for indications of cracks.

The magnetic particles tend to form on sharp corners and edges. Do **not** mistake these for cracks.

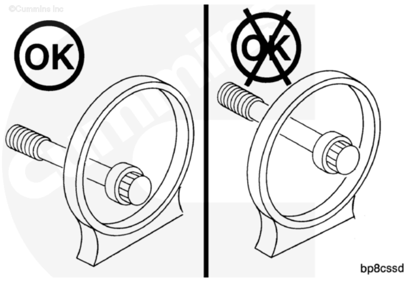
Any indications of cracks are **not** acceptable.




Prepare the machine for a coil shot.

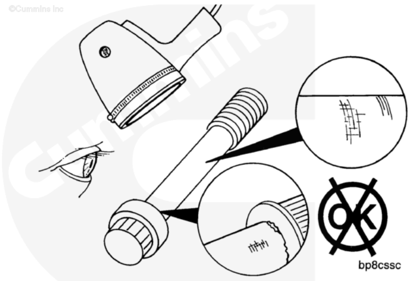
Verify the capscrew is near one side of the coil and **not** in the center.

Apply 1000 to 1350 ampere-turns.



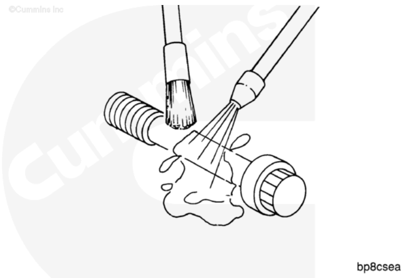

Use an ultraviolet light. Check for indications of cracks.

Any indications of cracks are **not** acceptable.




**⚠ CAUTION ⚠**

**The capscrew must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.**




Demagnetize the capscrew thoroughly.

Use solvent to clean the capscrew and dry with compressed air.

## Finishing Steps

**⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

**⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

**⚠ CAUTION ⚠**

**The lubricating oil system must be primed before operating the engine after any internal engine repairs to reduce the possibility of internal component damage.**

The piston and connecting rod **must** be installed as an assembly.

- Assemble and install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)
- Install the piston cooling nozzle. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.